

1. $P(5,5) * C(6, 3) * 1/2 = 1200$
2. $P(10; 3, 2, 2) + P(10; 2, 2, 2) + P(10; 3, 2, 2) + P(10; 3, 2, 2) = 2 * 10! / 8 = 907200$
3. (a) $P(8, 8) * 1/8 = 7! = 5040$
 (b) $P(7, 7) * 1/7 * 2 = 6! * 2/7 = 1440$
 (c) $P(6, 6) * 1/6 * P(6, 2) = 5! * 30 = 3600$
4. (a) $C(13, 1) * (52 - 4) = 624$
 (b) $C(4, 2) * [C(13, 2) * C(13, 3) + C(13, 1) * C(13, 4)] * 2 = 252824$
 (c) $C(52, 5) - 13^4 * 12 * C(4, 1) / 2 = 155252136$
 (d) $(52 * 3 / 2) * (48 * 3 / 2) / 2 * 44 = 123552$
5. $P(a + b + c; a, b, c) = (a + b + c)! / (a! * b! * c!)$
6. (a) 4^{12}
 (b) $[C(12, 3) * C(9, 3) * C(6, 3) * C(3, 3) / P(4, 4)] * P(4, 4) = 369600$
7. (a) $P(5, 1) * P(5, 1) * C(3, 1) = 75$
 (b) $2 * P(5, 1) * P(5, 1) * C(3, 1) - C(3, 2) * C(4, 1) * P(2, 2) = 150 - 24 = 126$
 (c) $3 * P(5, 1) * P(5, 1) * C(3, 1) + 2 * P(3, 3) - 3 * C(3, 2) * C(4, 1) * P(2, 2) = 165$
8. PROOF:

$$C(2n, 2) = 2n * (2n - 1) / 2 = 2 * n^2 - n$$

$$2 * (n, 2) + n^2 = n * (n - 1) + n^2 = 2 * n^2 - n$$

$$\Rightarrow C(2n, 2) = 2 * C(n, 2) + n^2$$

$$P(2n, 2) = 2n * (2n - 1) = 4n^2 - 2n$$

$$P(n, 2) = n * (n - 1) = n^2 - n$$

$$2 * P(n, 2) + n^2 = 3n^2 - 2n \neq P(2n, 2)$$

$$\Rightarrow C(2n, 2) = 2 * C(n, 2) + n^2$$

QED.

$$9. [P(2, 2) * 3 + P(2, 2) / P(2, 2)] + [C(2, 1) * C(2, 1) * C(1, 1) * 3 + P(3, 3) * 7 + P(3, 2) * 3 + P(3, 3) + C(2, 1) * C(2, 1) / P(2, 2)] + [C(2, 1) * P(3, 3) * 3 / P(2, 2) + C(3, 1) * P(3, 3) * 7 + C(3, 1) * P(3, 3) / P(2, 2) * 3 + P(3, 3) / P(2, 2) + P(4, 4) / P(2, 2) * 8 + P(4, 4) * 2 + P(4, 4) / P(3, 3) * 2 + P(4, 4) / P(2, 2) / P(2, 2) + 1 + C(2, 1) * P(3, 3) * 1 / P(2, 2) / P(2, 2)] = 415$$

$$10. (a) C(4, 1) * C(3, 1) * C(2, 2) / P(2, 2) * P(3, 3) = 36$$

$$(b) C(3, 2) = 3$$

$$(c) C(4, 1) * C(3, 1) * C(2, 2) / P(2, 2) = 6$$

$$(d) 1$$

$$11. (a) C(15, 3) = 455$$

$$(b) C(8, 3) = 56$$

$$(c) C(15, 3) - [C(5, 3) + C(6, 3) + C(9, 3) + C(10, 3) - C(12-5-6+4-1, 4-1)] = 225$$